Sahaj Bhadja

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Summary

New Graduate Software Engineer with 2+ years building **machine learning systems**, distributed services, data pipelines, and production tooling using Python, C++, PyTorch, Kubernetes, AWS, Docker, gRPC, automated testing, and performance optimization.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Web Technologies, Database Systems, Analysis of Algorithms, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Object-Oriented Programming, Operating Systems, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Architected a **multi-tenant ML platform** on Azure SQL and Blob Storage with indexed queries and workload isolation, reducing dashboard latency from 1.2 seconds to 350 milliseconds for 500+ users.
- Reworked asynchronous **LLM inference orchestration** with FastAPI workers, explicit job states, and failure isolation, tripling throughput for 1K+ concurrent jobs while sustaining 99%+ reliability across 5K+ monthly runs.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Built production **multimodal inference workflows** integrating GPT-4o, Claude, Qwen-VL, and OCR behind Flask APIs, supporting 150+ instructors and 10K+ daily requests with automatic model fallbacks reliably at scale.
- Designed concurrent ingestion for 1K+ files per batch using ThreadPoolExecutor, AWS RDS, and **CI/CD validation**, reducing peak-load feedback turnaround 60% while preserving responsive production experiences under load.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles, CA*

- Built a PyTorch **Decision Transformer** for autonomous-driving offline reinforcement learning, processing 100K+ trajectory logs through reproducible evaluation pipelines and improving simulated planning performance 18% across benchmark experiments.
- Profiled a spatial-temporal encoder combining HD-map geometry and ego trajectories through **multi-head attention**, improving closed-loop stability 15% while benchmarking latency, accuracy, and failure modes across simulation workloads.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Simplified distributed integration services through **object-oriented design patterns**, cutting API latency 30%, technical debt 20%, and duplicated logic while improving maintainability across a reliable cross-platform data-sharing system.
- Rebuilt AWS Glue and S3 **data pipelines** for incremental processing, improving efficiency 68%, then maintained 20+ automated test suites through Jenkins across Windows, Linux, and macOS environments.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*

- Built a **React and Node.js** operations platform with REST APIs, WebSockets, permissions, and notifications, synchronizing 2M+ records across locations while reducing staged-release conflicts 33% during production rollouts.
- Owned eight modular services from implementation through **Docker and Jenkins** deployment, adding automated validation and production monitoring that reduced release time 50% and sustained 95%+ workflow reliability.

Skills

Languages: Python, C++, Go, Java, TypeScript, JavaScript, SQL, Bash
ML & Serving: PyTorch, Transformers, ONNX, NVIDIA Triton, Ray, Embeddings, Recommendation Systems, Model Quantization
Distributed Systems: gRPC, REST APIs, Kafka, Redis, Async Processing, Dynamic Batching, Feature Stores, Service Mesh
Cloud & Data: AWS, EKS, S3, RDS, Kubernetes, Docker, PostgreSQL, Azure, Prometheus, Grafana
Engineering: Git, Linux, PyTest, CI/CD, GitHub Actions, Jenkins, Profiling, Benchmarking, Monitoring, On-Call, Unit Testing, Integration Testing

Projects

GPU Model Serving Platform | *Python, C++, PyTorch, ONNX, NVIDIA Triton, gRPC, EKS*

- Built a **GPU inference platform** that converts PyTorch models to ONNX, serves them through Triton and gRPC on EKS, and tunes dynamic batching to reduce latency 63% and compute cost 28%.

Streaming Feature Platform | *Python, Ray, Kafka, Redis, PostgreSQL, Kubernetes, Prometheus*

- Built a **distributed feature platform** with point-in-time joins, streaming updates, schema validation, and lineage tracking, enabling reproducible training datasets and low-latency online retrieval for recommendation models in production.

Big-Brain | *LLaMA 2, FAISS, Pinecone, PostgreSQL, AWS, REST APIs*

- Built an **embedding search and RAG system** using LLaMA 2, FAISS, Pinecone, and PostgreSQL, exposing scalable REST retrieval workflows and improving answer relevance through indexed semantic context.